The Independent T-test

The **t-test** assesses whether the means of two groups, or conditions, are statistically different from one other. They are reasonably powerful tests used on data that is parametric and normally distributed.

T-tests are useful for analysing simple experiments or when making simple comparisons between levels of your Independent Variable. There are two variants of the t-test:

- The **independent t-test** is used when you have <u>two</u> separate groups of individuals or cases in a *between-participants design* (for example: male vs female; experimental vs control group).

- The **repeated-measures t-test** (also known as the paired-samples or related t-test) is used when participants provide data for each level or condition of the independent variable in a *within-participants design* (for example, before and after an intervention).

For this tutorial we will focus on the **independent t-test**. There is another tutorial for the repeated-measures t-test. The reason for separating them is that the data files are setup differently, they use separate menu options in SPSS, and they produce different outputs.

The following example demonstrates what happens after you have created the data file. See *Tutorial 3: Adding Variables* for how to create your own file.

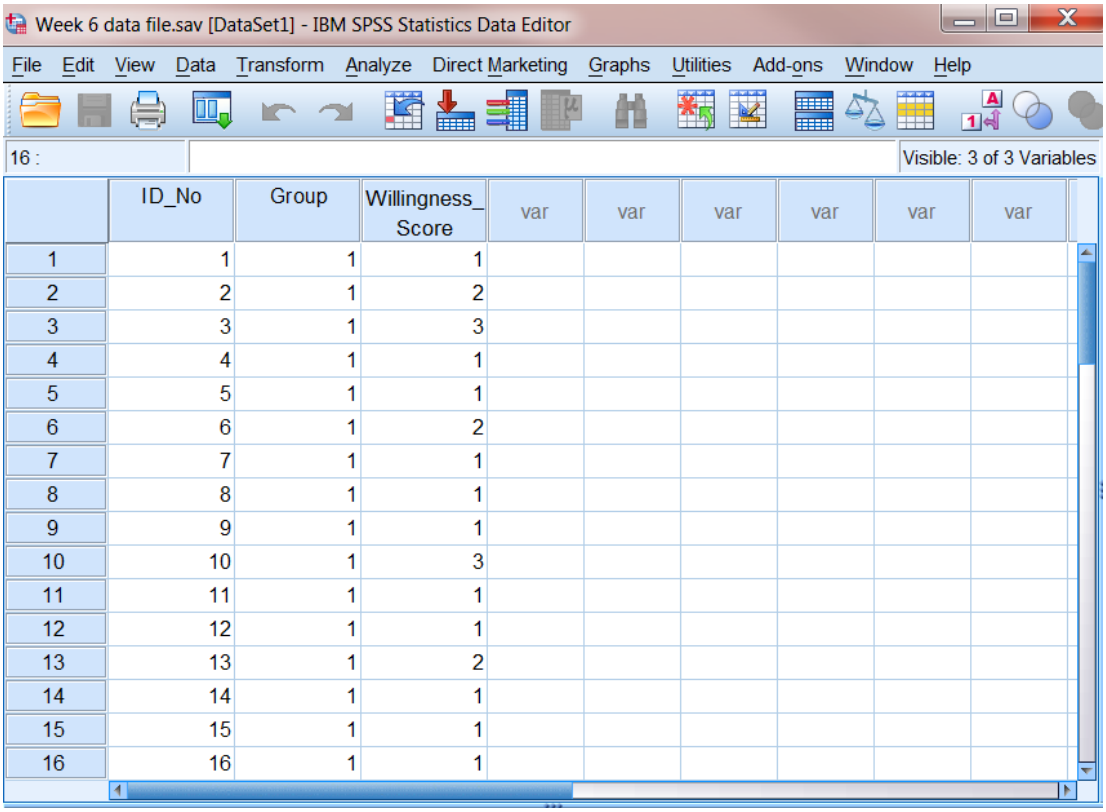
Using the Independent t-test in SPSS

This tutorial will walk you through how to run and interpret an Independent t-test. The example is based on a study by Shotland and Straw (1976), who were interested in how the perceived relationship between a couple fighting may affect the likelihood of somebody interventing.

To test this, participants were split into two group and asked to watch a video of two people fighting. The videos were identical, except for one crucial line. One group sees the victim shouting 'leave me alone, I don't know you', while the other group sees her saying 'leave me alone, I should never have married you'.

Participants were then asked to rate how willing they would be to intervene on a scale of 1-5.

The data can be found in the SPSS file: 'Week 6 data file.sav' and looks like this:



	ID_No	Group	Willingness_Score	var	var	var	var	var	var
1	1	1	1						
2	2	1	2						
3	3	1	3						
4	4	1	1						
5	5	1	1						
6	6	1	2						
7	7	1	1						
8	8	1	1						
9	9	1	1						
10	10	1	3						
11	11	1	1						
12	12	1	1						
13	13	1	2						
14	14	1	1						
15	15	1	1						
16	16	1	1						

For an independent t-test the data file should have *at least* two columns:

- one for the **independent variable**
- one for the **dependent variable**

The independent variable uses codes to assign group membership to each level or condition. For example, using '1' for a control group and '2' for an experimental group.
Each column represents a different variable, and each row contains the data from one participant. The different columns display the following data:

- The **ID_No** variable refers to the ID number assigned to each participant. We use numbers as identifiers instead of participant names, as this allows us to collect data while keeping the participants anonymous. This is good practice in psychology, especially when collecting potentially sensitive data.

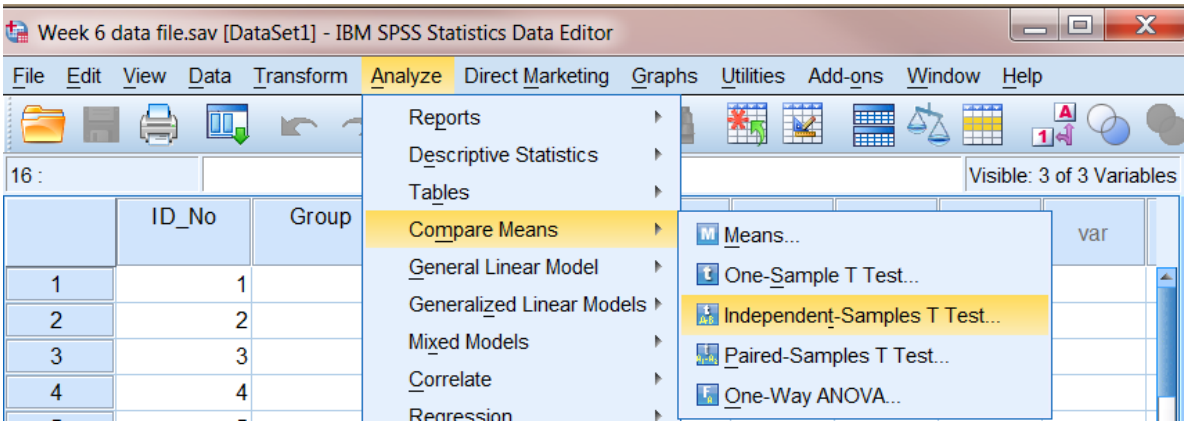
- The variable **Group** tells you which experimental condition participants are in. This is our *Independent Variable*. The IV for an Independent t-test will **always** be categorical (or nominal) data. It is easily recognisable by the fact that it uses category codes and labels.

  In this example, we have used the code '1' for participants who perceived the fighting couple as strangers, and '2' for those who perceived a relationship. These codes were defined in the SPSS Variable View screen. Refer to the earlier tutorial *Adding Variables* to see how this is done.

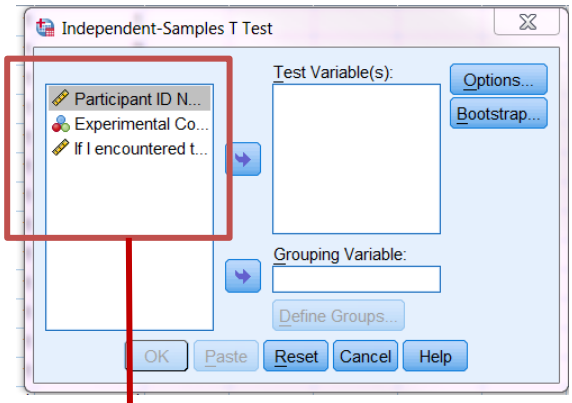
- **Willingness_Score** is a self-report rating of how willing participants would be to intervene in the fight they witnessed. This is our *Dependent Variable*. In our example this score is rated using a 5 point Likert scale, where higher scores indicate a higher likelihood of intervention).

As mentioned at the beginning of this tutorial, the independent t-test compares the scores of two groups on a certain variable. In this case, we want to compare the willingness scores of the two experimental groups.

To start the analysis, we first need to **CLICK** on the **Analyze** menu, select the **Compare Means** option, and then the **Independent-Samples T Test** sub-option.

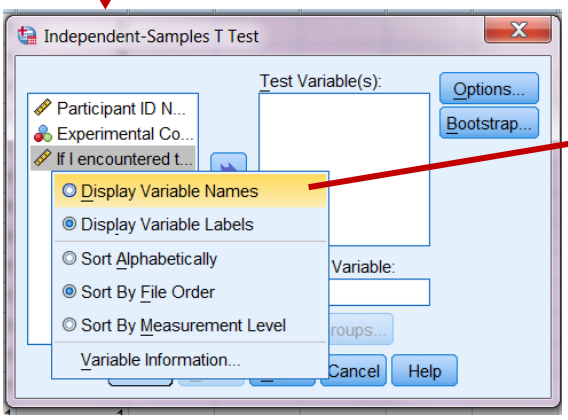


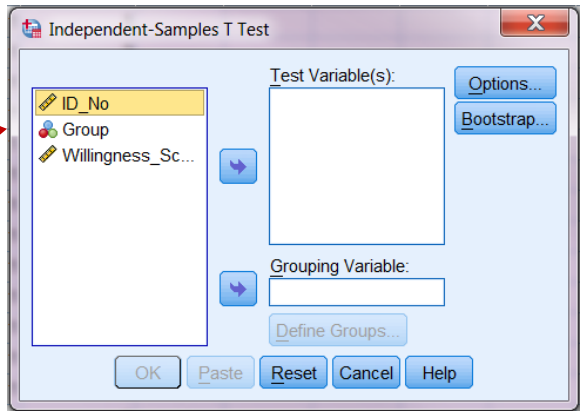
This opens up the **Independent Samples T-Test** dialog box. Here we need to tell SPSS which variables we want to analyse.



You may notice that your variables are now listed in the left hand window. As the Variable Labels are displayed, rather than the shorter Variable Names, they can be difficult to read.

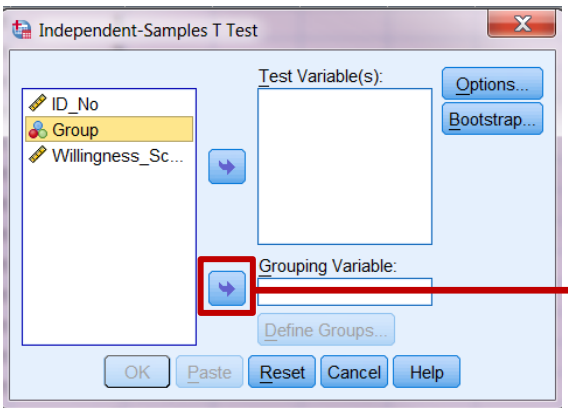
To change this, using your mouse to **<u>right click</u>** in the window and change the display option to 'Display Variable Names', as follows:

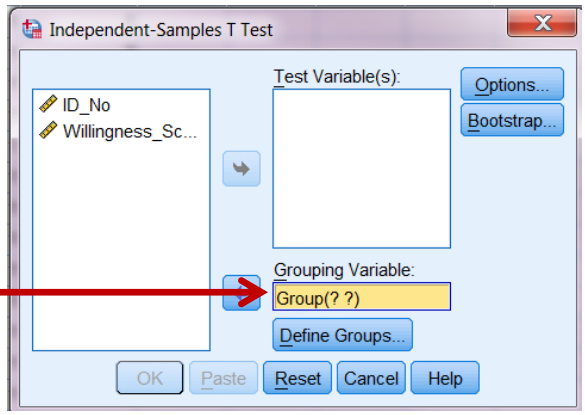




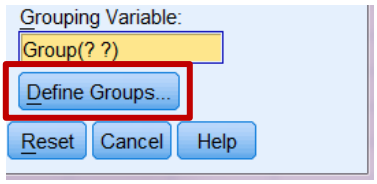
This changes the variable list so it is easier to read. You can now start the analysis.

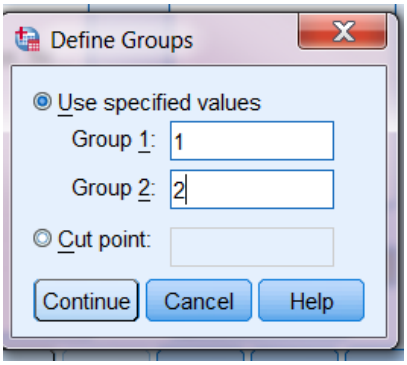
First, you need to tell SPSS what your **Grouping Variable** (or **IV**) is. To do this, **<u>SELECT</u>** the **Group** variable and move it across to the bottom right-hand pane using the blue arrow button next to the **Grouping Variable** box.





Group now has two question marks next to it. This means you have to tell SPSS which conditions you want to compare. As conditions of the IV (grouping categories) are entered into SPSS with numeric codes, you need to tell SPSS which codes represent the conditions you want to compare. You can do this by **CLICKING** on the **Define Groups** button.

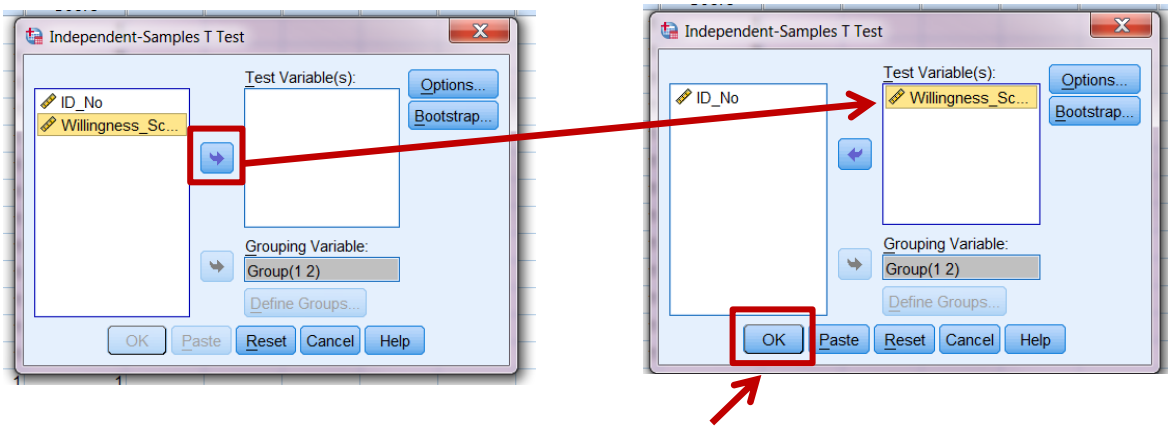




This opens the **Define Groups** dialog box, where you can enter the numeric codes for each experimental condition you are comparing. In this case, we are using '1' for perceived strangers and '2' for perceived relationship. As such, you need to add the numbers 1 and 2 to the Group 1 and Group 2 input boxes as illustrated here.

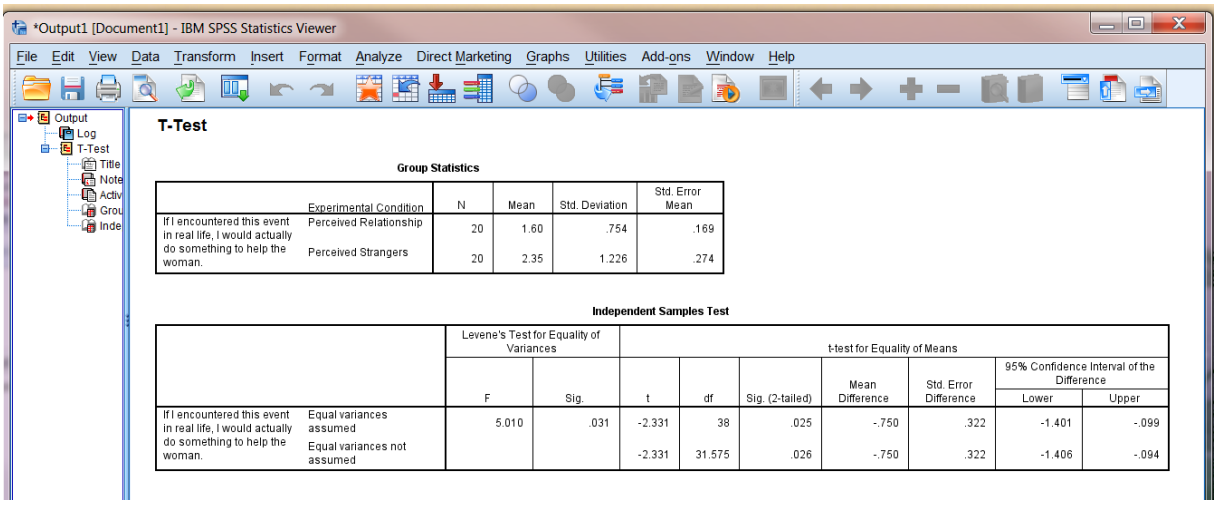
Once your have entered both numbers, the continue button should become active. **CLICK** on this to proceed.

You now need to tell SPSS what your dependent variable is by adding it to the analysis. To do this, select **Willingness_Score** and click on the upper arrow button to add it to the **Test Variable(s)** window.



Now that both variables have been added, **CLICK** on **OK** to run the analysis.

The output window allows you to inspect the results of the independent t-test:



Group Statistics

	Experimental Condition	N	Mean	Std. Deviation	Std. Error Mean
If I encountered this event in real life, I would actually do something to help the woman.	Perceived Relationship	20	1.60	.754	.169
	Perceived Strangers	20	2.35	1.226	.274

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
If I encountered this event in real life, I would actually do something to help the woman.	Equal variances assumed	5.010	.031	-2.331	38	.025	-.750	.322	-1.401	-.099
	Equal variances not assumed			-2.331	31.575	.026	-.750	.322	-1.406	-.094

Note that the output has two parts. These represent:

1.) the **descriptive statistics**...

2.) and the **inferential statistics**

Let's take a closer looks at the output boxes, one at a time.

Group Statistics

	Experimental Condition	N	Mean	Std. Deviation	Std. Error Mean
If I encountered this event in real life, I would actually do something to help the woman.	Perceived Relationship	20	1.60	.754	.169
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Group Statistics

The first box displays the **descriptive statistics** for your groups. It is always useful to inspect this box before you do anything else, as it allows you to gain initial insight into the pattern of your data.

In this table you can see that the mean willingness score for participants in the perceived relationship condition is 1.60, and 2.35 in the perceived strangers condition. In addition you can see from the standard deviations that the variation in the data (i.e. spread of scores) is a little wider for the strangers group (SD=1.23) than the relationship group (SD=0.75).

It is standard practice to report these descriptive statistics when reporting your results.

So by looking at your means you can see that, on average, participants who thought the fighting couple were in a relationship with one another were less likely to be willing to intervene than those who thought the couple were strangers.

But how should you interpret the difference between the means? To find out whether this observed difference between the scores is statistically significant, you next need to look at the table of inferential statistics.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
If I encountered this event in real life, I would actually do something to help the woman.	Equal variances assumed	5.010	.031	-2.331	38	.025	-.750	.322	-1.401	-.099
	Equal variances not assumed			-2.331	31.575	.026	-.750	.322	-1.406	-.094

Independent Samples Test

This box displays your **inferential statistics**: the output from the **independent t-test**. You don't need to worry about all of the columns here (many parts of the table are only needed at a more advanced level). The key sections of the table are highlighted above and described below:

- **Levene's Test of Equality of Variances :** An assumption of the independent t-test is that the two groups you are comparing have a similar dispersion of scores (otherwise known as homogeneity or equality of variance). These columns tell us whether or not this is the case. If the value of *F* is significant, this indicates that there are statistically significant differences in the way the data are dispersed, and the assumption of homogeneity has ***not*** been met.

  Note that the output table has two rows: we use one when variances are equal and the other when they are not. In this example our variances *cannot* be assumed to be equal as the *F*-value is significant (p = .031). As such, we only need to read the values from the ***second row*** of the table.

- **t-test for Equality of Means :** This column is where the statistics for the t-test are found. This section is divided into seven sub-sections, but only three of the columns are important at this stage. They are: labelled *t*, *df* and *sig (2-tailed)*.

  This is where you determine whether or not there is support for the hypothesis tested. **CLICK** on these subheadings to find out more.

- **t - Obtained Value of t :** This is the value of *t* -test statistic that SPSS has calculated. The larger the value of *t*, the smaller the probability that the results occurred by chance.

  Before computers were used for data analysis, you would have had to calculate the value of *t* by hand using a formula. The obtained value would then be compared to something called a *critical value* in a table known as the *t*-distribution. SPSS saves us the job of having to do this calculation... *phew*!!

- **df - Degrees of Freedom :** You will come across degrees of freedom in most statistical tests. It is a value we use to represent the size of the sample or samples used in a statistical test and it needs to be reported.

  The way that degrees of freedom are calculated varies for different statistical tests, but they must be calculated correctly before a test result can be checked for significance. Don't worry too much about this, as SPSS automatically calculates the value for you… but it might be useful to note that with independent *t*-tests the *df* is always close to the total number of participants.

- **Sig (2-tailed) :** The significance level (also called the probability or *p*-value) tells us the likelihood that our results have occurred by chance. If this value is smaller than .05 then there is support for our hypothesis. If it is larger, then we reject our hypothesis in favour of the null hypothesis... which is that there are no differences between the two groups.

  For an independent t-test, SPSS reports the test at a 2-tailed significance level by default. To obtain a one-tailed probability (when your hypothesis is *directional*) simply divide the *p*-value in half. In this case, we could test a one-tailed hypothesis that people who perceive a relationship between a fighting couple will be less willing to intervene in a fight than those who do not, as this is what the literature suggests. If we were to do this, our *p*-value would be 0.013 (1-tailed).

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
If I encountered this event in real life, I would actually do something to help the woman.	Equal variances assumed	5.010	.031	-2.331	38	.025	-.750	.322	-1.401	-.099
	Equal variances not assumed			-2.331	31.575	.026	-.750	.322	-1.406	-.094

So what do we need to know from our output?

When writing up the results of your *t*-test you need to report whether or not the test was significant following this formula:

t (df) = t value, p = p value

...where you insert the relevant numbers into the underlined sections. For this particular example, we have found that the *t*-test *is* significant as the p-value is less than 0.05 (p< .05). This is reported as:

$t(31.58) = -2.33$, $p = 0.026$

What do our findings tell us?

When interpreting and writing up your findings you need to use information from *both* the **descriptive** and the **inferential** statistics in your output. The order in which you present these two types of statistics doesn't really matter, but always finish by clearly interpreting your results.

- **Step One:** You can *describe* the pattern of your data using the means and standard deviations from the first output table. In this case you could say something like:

  Results showed participants who saw a relationship between the couple had lower willingness scores (M=1.60, SD=.75) than those who did not (M=2.35, SD=1.22).

- **Step Two:** Use both words and numbers to formally report whether or not this difference is *significant*:

  An independent t-test found this pattern to be significant, t(31.58) = -2.33, p < 0.05.

- **Step Three:** Finally, you need to put this information together to interpret and summarise what you have found in terms of your hypothesis. This should be written *in plain English*, for example:

Together this suggests the perceived relationship the perceived relationship between the victim and perpetrator affects participants' willingness to intervene, supporting our hypothesis.

What next…

Now you have learned how to carry out an Independent t-test using SPSS, why not try downloading the Week 6 data file and see if you can produce the same output on your own?

Remember, practice makes perfect.